Bonus Assignment 2 - FrameLab: User Guide

November 8, 2026 | Badr Elsayed

1. Accessing the tool

FrameLab is a browser-based learning tool for **2D and 3D beam/frame FEM**. Open the live tool in any current browser: **https://badrelsayednyu.github.io/beam-frame-fem/**
No installation or login is required. All calculations and explanations run in the browser.

Source repository: github.com/badrelsayedNYU/beam-frame-fem. Download **index.html** and open it locally for completely offline use; no external library is needed. The working HTML app and source ZIP are also embedded in this PDF as backup attachments. The live link above is the simplest way to open the tool.

Unlike a truss tool, which models axial bar forces, this assignment also requires bending and nodal rotations, plus torsion and bending about two axes in 3D. These additional effects are included in the seven-stage walkthrough.

2. Operating the tool

Choose a built-in case from **Start with an example**, then use the stage bar or **Previous / Next step** buttons. Edit the model in the left panel and select **Solve & explore** to refresh all calculations.

Stage	What it shows	What you can do
1 Model & DOFs	Node order, global DOF numbering, translations/rotations, restraints, and model drawing.	Edit node coordinates, supports, loads, and member connections.
2 Element stiffness	Member length, axial/bending/torsional contributions, shape functions, coefficients, and local stiffness.	Select a member and isolate an individual physical contribution.
3 Transform	Local/global axes, rotation R, transformation T, and $k_{global} = T^T k_{local} T$.	Inspect the transformation and global stiffness of each member.
4 Assembly	Local-to-global DOF mapping, selected member contribution, and complete global K.	Choose a row/column to see the sum of member contributions.
5 Constraints	Global loads, free/constrained DOFs, prescribed values, reduced Kff, and adjusted RHS.	Inspect how supports and prescribed movements alter the equations.
6 Solve	Diagonal scaling, Gaussian elimination, row operations, and back substitution.	Move the elimination slider to inspect intermediate matrices.
7 Results	Nodal translations/rotations, reactions, residuals, and local member end forces.	Inspect a member and compare original and deformed structures.

Drawing controls. Toggle the deformed shape and set its magnification. In 3D, use Orbit and Elevation to change the view. Click a member to inspect it. Hover nodes for coordinates/DOFs and supports for prescribed movements.

FrameLab: models and demonstration examples

Defining your own problem. In Nodes & supports, add/edit coordinates and check each prescribed DOF. Fixed restrains all DOFs; Pinned restrains translations; Free releases all. Individual controls also allow rollers and nonzero prescribed movements. In Members, choose endpoint nodes and enter positive section/material properties. In Loads, enter nodal forces and moments along global axes. There is no fixed node/member limit; browser resources determine practical size.

Units. Use one consistent system: E and G are force/length^2, A is length^2, I and J are length^4, and moments are force x length. Translations use the length unit; nodal rotations are radians; section roll is degrees. Unit labels do not convert values. Positive rotations follow the right-hand rule. Save model/Open model export and restore JSON. Invalid inputs and unstable or insufficiently restrained structures produce an error.

3. Demonstration examples: 2D and 3D

2D portal frame - supplied textbook Example 5.1. Four nodes at (-60,0), (-60,120), (60,120), (60,0) in inches form two columns and a top beam. Both bases are fixed. For every member, E = 30 x 10^6 psi and A = 10 in^2; column Iz = 200 in^4, beam Iz = 100 in^4. Node 2 has Fx = 10,000 lb; node 3 has Mz = 5,000 lb-in.

Walkthrough. Three 6 x 6 element matrices transform and assemble into a 12 x 12 K. The six fixed base DOFs leave a 6 x 6 reduced system. Solve for top-joint movements, then recover reactions from the original equations.

Quantity	FrameLab	Textbook (rounded)
Node 2 ux (in)	0.211363	0.211
Node 2 uy (in)	0.00148133	0.00148
Node 2 rz (rad)	-0.00152603	-0.00153
Node 3 ux (in)	0.209359	0.209
Node 3 uy (in)	-0.00148133	-0.00148
Node 3 rz (rad)	-0.00148600	-0.00149

Differences reflect rounded textbook stiffness coefficients. Global support reactions balance the applied force and moment. The app retains unrounded coefficients during computation.

3D combined-load cantilever. A 3 m member runs from fixed (0,0,0) to free (3,0,0), with zero section roll. E = 210 GPa, G = 80 GPa, A = 0.01 m^2, Iy = 8 x 10^{-6} m^4, Iz = 6 x 10^{-6} m^4, J = 10^{-5} m^4. Tip forces are [10000,-1000,-1500] N and moments [600,0,0] N-m. One 12 x 12 element leaves six free DOFs after fixing the base.

Response	Independent analytical expression	FrameLab
ux (m)	Fx L / (EA)	1.42857e-5
uy (m)	Fy L^3 / (3 E Iz)	-0.00714286
uz (m)	Fz L^3 / (3 E Iy)	-0.00803571
rx (rad)	Mx L / (GJ)	0.00225
ry (rad)	-Fz L^2 / (2 E Iy)	0.00401786
rz (rad)	Fy L^2 / (2 E Iz)	-0.00357143

All six responses agree with the analytical expressions. Fixed-base reactions are [-10000,1000,1500] N and [-600,-4500,3000] N-m. A third built-in case, a three-member 3D bent frame, demonstrates spatial orientation and multi-member assembly.

FrameLab: source, validation, and disclosure

4. Source files and how the computation works

The GitHub repository follows the earlier truss assignment format: a self-contained **index.html** and **README.md** at the root. Readable modules are **src/fem.js** (mathematical engine), **src/app.js** (editor, explanations, matrices, and drawing), and **src/style.css**. Tests, this guide, a validation log, and AI disclosure are included. No build step is required. The embedded backup ZIP contains the same app using the original **dist/** layout.

Element formulation. A 2D node has [ux,uy,rz]; a 3D node has [ux,uy,uz,rx,ry,rz]. The member matrix combines axial EA/L, Euler-Bernoulli bending terms $12EI/L^3$, $6EI/L^2$, 4EI/L, and 2EI/L, plus 3D torsion GJ/L. Cubic Hermite bending functions interpolate displacements and end rotations. The app shows the derivation, coefficients, and separate contributions.

Transformation and assembly. The **element(model,e)** function computes length, local axes, local stiffness, T, transformed stiffness, and DOF mapping. Local x points from the first endpoint to the second. In 3D, a stable reference vector constructs the remaining axes; section roll rotates the y/z axes about x. Each transformed matrix is added at its global DOF positions.

```
u_local = T u_global
k_global = T^T k_local T
```

Constraints and solution. The **solve(model)** function assembles K and F, partitions free and prescribed DOFs, and includes nonzero prescribed movements in the reduced right-hand side. It uses diagonal scaling followed by Gaussian elimination with partial pivoting and back substitution. Stored operations let Stage 6 reconstruct each intermediate system.

```
Kff uf = Ff - Kfc uc
R = K u - F
f_element_local = k_local T u_element
```

Reactions use the original unreduced K and applied load vector. Free-DOF residuals should be close to zero. The **curve(element,dimension,t)** function uses linear axial and cubic bending interpolation for the displayed centerline. Deformation magnification changes the picture only.

Validation. Run **node tests/fem.test.cjs** with Node.js. All 15 test groups passed, including the supplied 2D benchmark, six analytical 3D responses, force/moment equilibrium, stiffness symmetry, arbitrary orientation/roll and energy invariance, rigid motions, prescribed displacement, interpolation, split-member equivalence, singularity detection, invalid inputs, and reproducibility. Browser checks covered the seven stages, model editing, support controls, error recovery, and the self-contained app.

Scope. Linear elastic, small-displacement, prismatic Euler-Bernoulli members with rigid joints and nodal loading. Internal member hinges/releases, distributed member loads, tapered sections, shear deformation, buckling, and nonlinear response are not implemented. The supplied second test PDF contains several special beam exercises without a marked target; those special formulations are not claimed as validated cases. No worked 3D benchmark was supplied, so the analytical cantilever supplies independent 3D verification.

5. AI usage disclosure

OpenAI Codex assisted with interpreting the assignment, writing the HTML/CSS/JavaScript source and FEM engine, preparing the interface and mathematical explanations, drafting documentation, and developing validation tests. Numerical results were checked against the supplied portal-frame reference, analytical cantilever expressions, and equilibrium/invariance checks.

The finished tool does **not** call an AI model, chatbot, LLM, or external AI service during operation. Every stiffness matrix, transformation, displacement, reaction, member force, and explanation is generated locally by implemented equations, algorithms, and fixed templates. Identical inputs produce consistent, reproducible results.